DASH-04: Operations Dashboards

> **Series:** DASH | **Notebook:** 4 of 7 | **Created:** March 2026 | **Last Updated:** 04/04/2026

Overview

Operations dashboards serve as the real-time control panel for SREs, platform engineers, and NOC teams. Unlike executive dashboards that summarize weekly trends, operations dashboards focus on what is happening right now — service response times, error rates, log volume anomalies, active problems, and infrastructure health. This notebook covers tile selection, query patterns for real-time monitoring, and layout strategies for operations-tier dashboards.

Table of Contents

1. Service Response Time Monitoring
2. Error Rate by Service
3. Log Volume and Anomaly Tracking
4. Active Problems and Alerts
5. Infrastructure Health
6. Deployment Correlation
7. Operations Dashboard Layout
8. Summary and Next Steps

Prerequisites

Requirement	Details
Dynatrace Environment	SaaS or Managed with Grail enabled
Permissions	`storage:logs:read` , `storage:metrics:read` , `storage:events:read` , `storage:spans:read`
Data	Active services with span data, log ingestion, host metrics
Prior Reading	DASH-01, DASH-02

1. Service Response Time Monitoring

Response time is the most watched metric on an operations dashboard. Show it as a timeseries line chart with p50, p90, and p95 lines for context.

Response Time Trend (All Services)

```
// Service response time trend — p50, p90, p95 over last 2 hours
fetch spans, from:-2h
| filter span.kind == "server"
| makeTimeseries p50_ms = percentile(duration, 50) / 1000000, p90_ms =
percentile(duration, 90) / 1000000, p95_ms = percentile(duration, 95) /
1000000, interval:5m
```

Response Time by Service (Top 10 Slowest)

```
// Top 10 slowest services by average response time
fetch spans, from:-1h
| filter span.kind == "server"
| summarize avg_ms = avg(duration) / 1000000, p95_ms = percentile(duration,
95) / 1000000, request_count = count(), by:{dt.entity.service}
| sort p95_ms desc
| limit 10
```

2. Error Rate by Service

Error rate as a percentage shows which services are degraded. Display as a bar chart sorted by error rate descending.

Current Error Rate Table

```
// Error rate by service — operations bar chart
fetch spans, from:-1h
| filter span.kind == "server"
| summarize total = count(), errors = countIf(otel.status_code == "ERROR"),
by:{dt.entity.service}
| fieldsAdd error_rate_pct = round(100.0 * errors / total, decimals: 2)
| sort error_rate_pct desc
| limit 15
```

Error Rate Trend Over Time

```
// Overall error rate trend — operations line chart
fetch spans, from:-2h
| filter span.kind == "server"
| makeTimeseries total = count(), errors = countIf(otel.status_code ==
"ERROR"), interval:5m
| fieldsAdd error_rate = arrayAvg(errors) / arrayAvg(total) * 100
```

3. Log Volume and Anomaly Tracking

Sudden log volume changes often indicate problems before Davis detects them. Track log volume by level and source.

Log Volume by Severity Level

```
// Log volume by severity — stacked area chart
fetch logs, from:-2h
| filterOut loglevel == "NONE"
| makeTimeseries log_count = count(), interval:5m, by:{loglevel}
```

Top Log Sources by Error Volume

```
// Top 10 log sources generating errors — operations table
fetch logs, from:-1h
| filter loglevel == "ERROR"
| summarize error_count = count(), by:{log.source}
| sort error_count desc
| limit 10
```

Log Volume Spike Detection

Compare current hour to the same hour yesterday to detect anomalous volume increases.

```
// Log volume comparison: current hour vs same hour yesterday
fetch logs, from:-1h
| summarize current_count = count()
| append [
    fetch logs, from:now() - 25h, to:now() - 24h
    | summarize yesterday_count = count()
  ]
```

4. Active Problems and Alerts

Active problems should be front and center on every operations dashboard. Show both the count and the detail.

Active Problems List

```
// Active problems detail table — operations dashboard
fetch dt.davis.problems, from:-24h
| filter event.status == "ACTIVE"
| filter dt.davis.is_duplicate == false
| fieldsAdd duration_min = (toLong(now()) - toLong(event.start)) /
60000000000.0
| fieldsKeep display_id, event.name, event.category, duration_min
| sort duration_min desc
```

Problem Status Over Time

```
// Concurrently open problems over time — area chart
fetch dt.davis.problems, from:-24h
| makeTimeseries concurrent = count(), spread:timeframe(from:event.start,
to:coalesce(event.end, now()))
```

5. Infrastructure Health

Infrastructure tiles provide context for service-level issues. CPU, memory, and disk metrics help operations teams determine if a problem is service-specific or infrastructure-wide.

Host CPU Usage

```
// Host CPU usage — operations line chart
timeseries avg_cpu = avg(dt.host.cpu.usage), from:-2h, by:{dt.entity.host}
```

Top Hosts by CPU — Identifying Hot Spots

```
// Top 5 hosts by average CPU usage
timeseries avg_cpu = avg(dt.host.cpu.usage), from:-1h, by:{dt.entity.host}
| fieldsAdd avg_cpu_val = arrayAvg(avg_cpu)
| sort avg_cpu_val desc
| limit 5
```

Host Memory Usage

```
// Host memory usage — operations line chart
timeseries avg_mem = avg(dt.host.memory.usage), from:-2h, by:{dt.entity.host}
```

6. Deployment Correlation

Deployments are the most common cause of production issues. Overlaying deployment events on service metrics helps operations teams immediately correlate "what changed."

Recent Deployment Events

```
// Recent deployment events — operations context table
fetch events, from:-6h
| filter event.kind == "DEPLOYMENT_EVENT" or event.type ==
"CUSTOM_DEPLOYMENT"
| fieldsKeep timestamp, event.type, dt.entity.service, event.name
| sort timestamp desc
| limit 20
```

7. Operations Dashboard Layout

Recommended Tile Arrangement

Section	Tiles	Chart Type
Status Bar (top)	Active problems count, overall error rate, log error count	Single values
Service Health	Response time trend, error rate by service	Line chart, bar chart
Log Intelligence	Log volume by level, top error sources	Stacked area, table
Infrastructure	CPU by host, memory by host	Line charts
Change Context	Recent deployments	Table

Operations Dashboard Anti-Patterns

Anti-Pattern	Problem	Fix
Too many services on one chart	Unreadable spaghetti lines	Use variables or top-N filtering
All tiles show 24h data	Hides real-time spikes	Use 1-2h for real-time tiles

Anti-Pattern	Problem	Fix
No problem context	Team sees metrics but not root cause	Add active problems table
Missing deployment markers	Cannot correlate changes with incidents	Add deployment events section

8. Summary and Next Steps

In this notebook you learned:

- How to build service response time and error rate tiles for real-time monitoring
- Log volume tracking and spike detection queries
- Active problem tables with duration tracking
- Infrastructure health tiles for CPU and memory
- Deployment event tracking for change correlation
- Recommended layout and common anti-patterns for operations dashboards

Next: In **DASH-05: Engineering Dashboards**, we move to the deepest tier — trace-level analysis, database performance, endpoint metrics, and before/after deployment comparisons.
